

Project Report

“Network Intrusion Detection using ANN on NSL-KDD Dataset”

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Abstract:

In the Area of Security, Intrusion Detection System (IDS) form an individual trailing and plays an essential role in information Security. As the usability of the internet among the users in a wide area is increasing day by day so as the importance of security and to keep the system aware of the malicious activities is also increasing. This project explores the application of machine learning techniques for network intrusion detection using the NSL-KDD dataset. We pre process the data, perform feature extraction, and utilize an Artificial Neural Network (ANN) model to classify network traffic into different attack categories. The results demonstrate the potential of machine learning in effectively identifying intrusions within network environments. Key steps include data loading and pre processing, feature engineering using domain knowledge, application of SMOTE for data imbalance, ANN model training and validation, performance evaluation of model using metrics like accuracy, validation loss and visualization by confusion matrix. The model shows 99.65% test accuracy, indicating the efficiency of machine learning techniques in classifying network attacks.

Introduction:

Intrusion is a set of actions that attempts to compromise the integrity, confidentiality, or availability of any resource on a computing platform. An intrusion detection system (IDS) is a system that monitors network traffic for doubtful activity and matters alert when such activity is exposed. Network security is crucial for maintaining the integrity and confidentiality of computer systems. Intrusion Detection Systems (IDS) play a vital role in identifying malicious activities within networks. Feature selection for intrusion detection is most important factor for the success of intrusion detection system. The objectives of IDS are confidentiality, integrity, availability and 1accountability. Traditional signature-based IDS often struggle with novel attacks, highlighting the need for adaptive and intelligent methods. Machine learning offers a promising alternative by enabling systems to learn patterns from data and detect anomalies. This project investigates the effectiveness of an Artificial Neural Network (ANN) in detecting network intrusions using the NSL-KDD dataset, a commonly used benchmark in the field.

Dataset Description:

The statistical analysis showed that there are important issues in the KDD data set which highly affects the performance of the systems, and results in a very poor estimation of anomaly detection approaches. To solve these issues, a new data set as, NSL-KDD is used, which consists of selected records of the complete KDD data set.

The advantage of NSL KDD dataset are:

1. No redundant records in the train set, so the classifier will not produce any biased result
2. No duplicate record in the test set which have better reduction rates.
3. The number of selected records from each difficult level group is inversely proportional to the percentage of records in the original KDD data set.

The training dataset is made up of 21 different attacks out of the 37 present in the test dataset. The known attack types are those present in the training dataset while the novel attacks are the additional attacks in the test dataset i.e. not available in the training datasets. The attack types are grouped into four categories: DoS, Probe, Privilege and Remote Access

Literature Review:

The NSL-KDD dataset is a refined version of the original KDD Cup 1999 dataset [1], addressing some of the inherent issues like redundancy. Previous research using this dataset has shown the effectiveness of various machine learning algorithms including Support Vector Machines (SVM), Random Forests, Naive Bayes classifiers, and now deep learning models including ANNs. A detailed analysis on the NSL-KDD data set using various machine learning techniques is done in [4] available in the WEKA tool. K-means clustering algorithm uses the NSL-KDD data set [5] to train and test various existing and new attacks. A comparative study on the NSL-KDD data set with its predecessor KDD99 cup data set is made in [6] by employing the Self Organization Map

(SOM) Artificial Neural Network. An exhaustive analysis on various data sets like KDD99, GureKDD and NSLKDD are made in using various data mining based machine learning algorithms like Support Vector Machine (SVM), Decision Tree, K-nearest neighbor, K-Means and Fuzzy C-Mean clustering algorithms. These approaches have achieved varying degrees of accuracy in classifying network traffic. Recent works explore the use of feature selection and engineering techniques to enhance the performance of these models, and to address imbalanced data using oversampling techniques like SMOTE. Prior studies in network anomaly detection focus on different aspects like traffic features, time series analysis, and hybrid classification models. This project builds upon these foundations by exploring an ANN approach after careful data preprocessing and applying oversampling techniques to address data imbalance issues in NSL-KDD dataset.

Methodology:

This project's methodology consists of the following key steps:

1. Data Loading and Preparation:

- The NSL-KDD datasets are loaded using pandas.
- Column names are assigned to ensure clear understanding of the data.
- The training and testing datasets are merged into a single data frame to enable consistent processing.
- Missing values are checked and verified to be absent in our dataset.

2. Data Preprocessing:

- The dataset's 'attack' column, containing specific attack types, was mapped to a simpler set of 5 classifications using a custom mapping function: 'normal', 'dos', 'probe', 'privilege', and 'remote_access'. This simplified the prediction problem from a multi-class to a 5 class problem. The distribution of attack_map is:

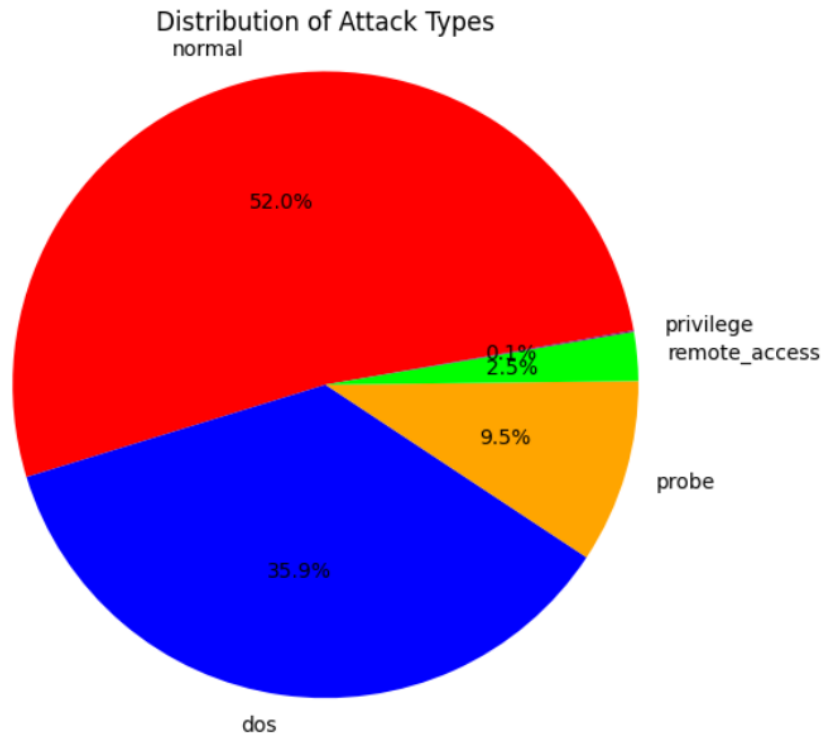


fig1.1

The original 'attack' column is then removed to avoid redundancy.

- String-based categorical features like 'protocol_type', 'service', and 'flag' are converted into numerical representations using Label Encoding. Also the encoded labels are displayed that refers to its original labels.

3. Feature Selection:

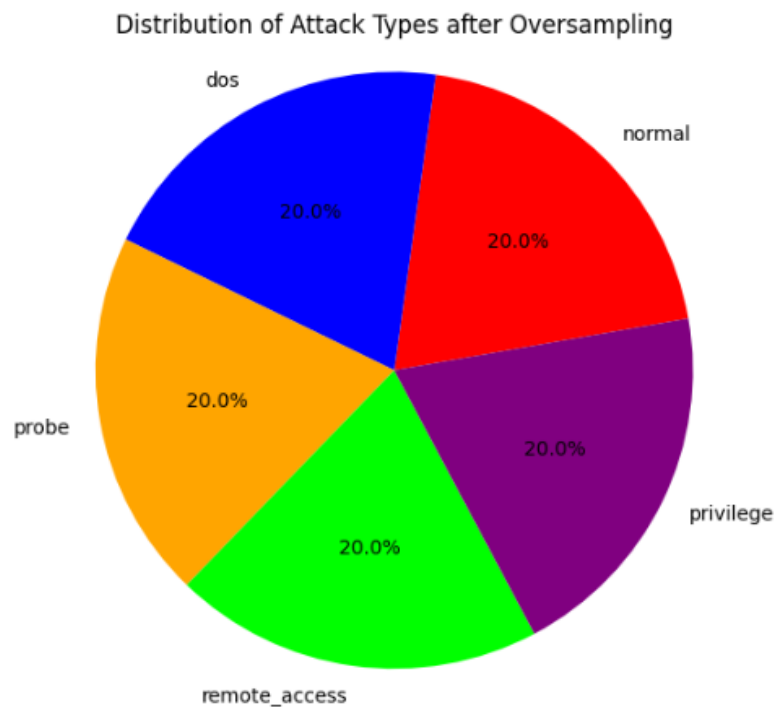
- In this study, feature selection was performed to address **multicollinearity** among the independent variables, ensuring that redundant features were eliminated to enhance model interpretability and predictive performance. The analysis began with the computation of the correlation matrix for all features using the Pearson correlation coefficient. A heatmap of the correlation matrix was visualized using Seaborn, highlighting feature pairs with high correlations (above 0.75 but not equal to 1.0), which indicated significant multicollinearity.

exhibited correlation coefficients below the predefined threshold of 0.75, validating the effectiveness of the feature selection process.

- By mitigating multicollinearity, this approach ensures that the predictive model developed in subsequent phases leverages features that contribute independently to the target variable. This not only enhances the robustness of the model but also provides clearer insights into the relationship between features and the target variable, fostering better decision-making in applied contexts.

4. Addressing Data Imbalance:

- The distribution of attack types is visualized using a pie chart as in fig1.1, revealing the class imbalance in the dataset.
- **SMOTE** (Synthetic Minority Oversampling Technique) is applied to the dataset to create balanced dataset of the same size. This helps avoid over fitting and improve model performance on under-represented classes.



5. Train-Validation-Test Split:

- The dataset is split into training, validation, and testing sets with a 70/18/12 split ratio respectively to ensure robust model evaluation.

6. Data Scaling:

- The numerical features are scaled using Standard Scalar for better model convergence.

7. Artificial Neural Network (ANN) Model:

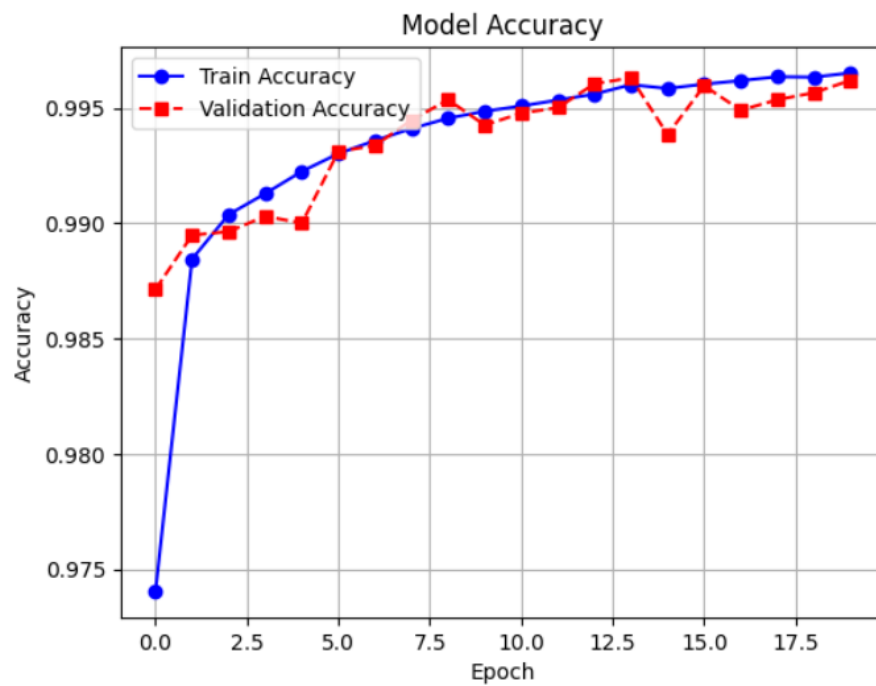
- A sequential ANN model is built with the following architecture:
 - a) An input layer with 64 neurons and 'relu' activation function.
 - b) A hidden layer with 32 neurons and 'relu' activation function.
 - c) An output layer with 5 neurons and 'softmax' activation function to provide class probabilities.
- The model is compiled using the 'adam' optimizer, 'sparse_categorical_crossentropy' loss function (suitable for integer encoded target), and 'accuracy' as a metric.

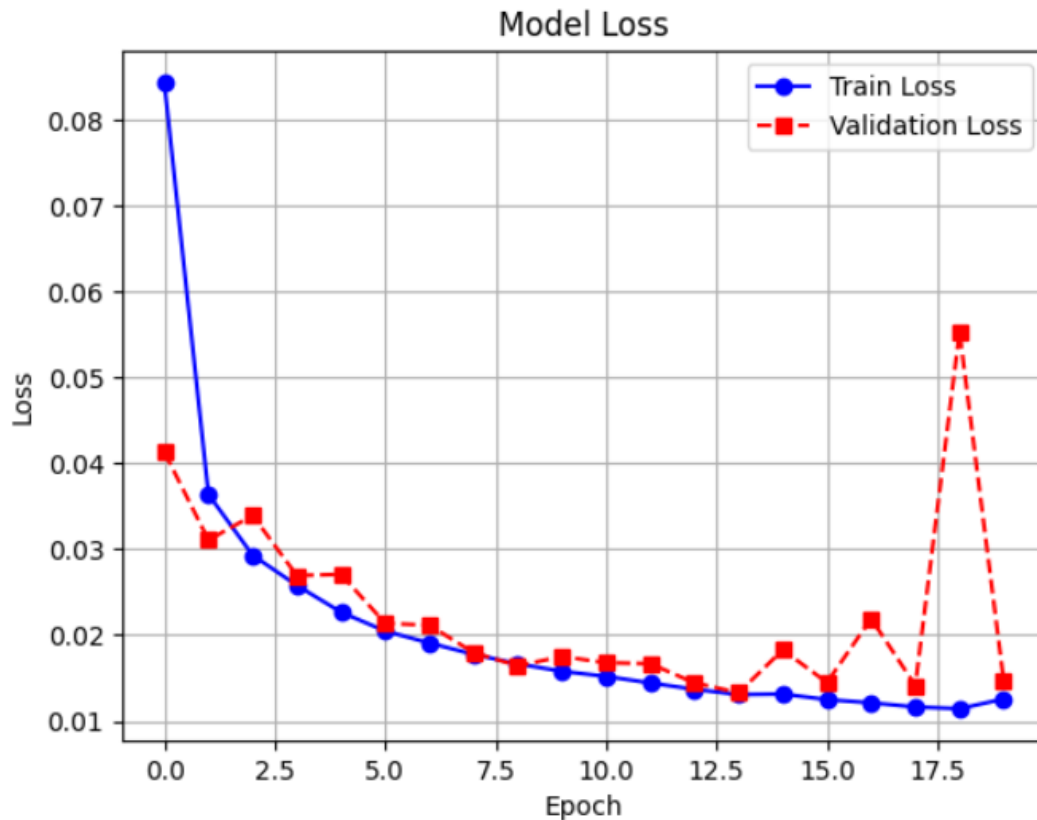
8. Model Training and Evaluation:

- The model is trained for 20 epochs using training data.
- Validation data is provided to monitor performance during training.
- Model performance is evaluated on the validation set using the validation accuracy and validation loss.
- Finally, the trained model is evaluated on the test data set to determine the test accuracy and loss.

Results

The training and validation accuracy and loss are plotted for each epoch to assess the model performance trend. A confusion matrix is generated and visualized using Seaborn to analyse the classification errors of the model on the test set.

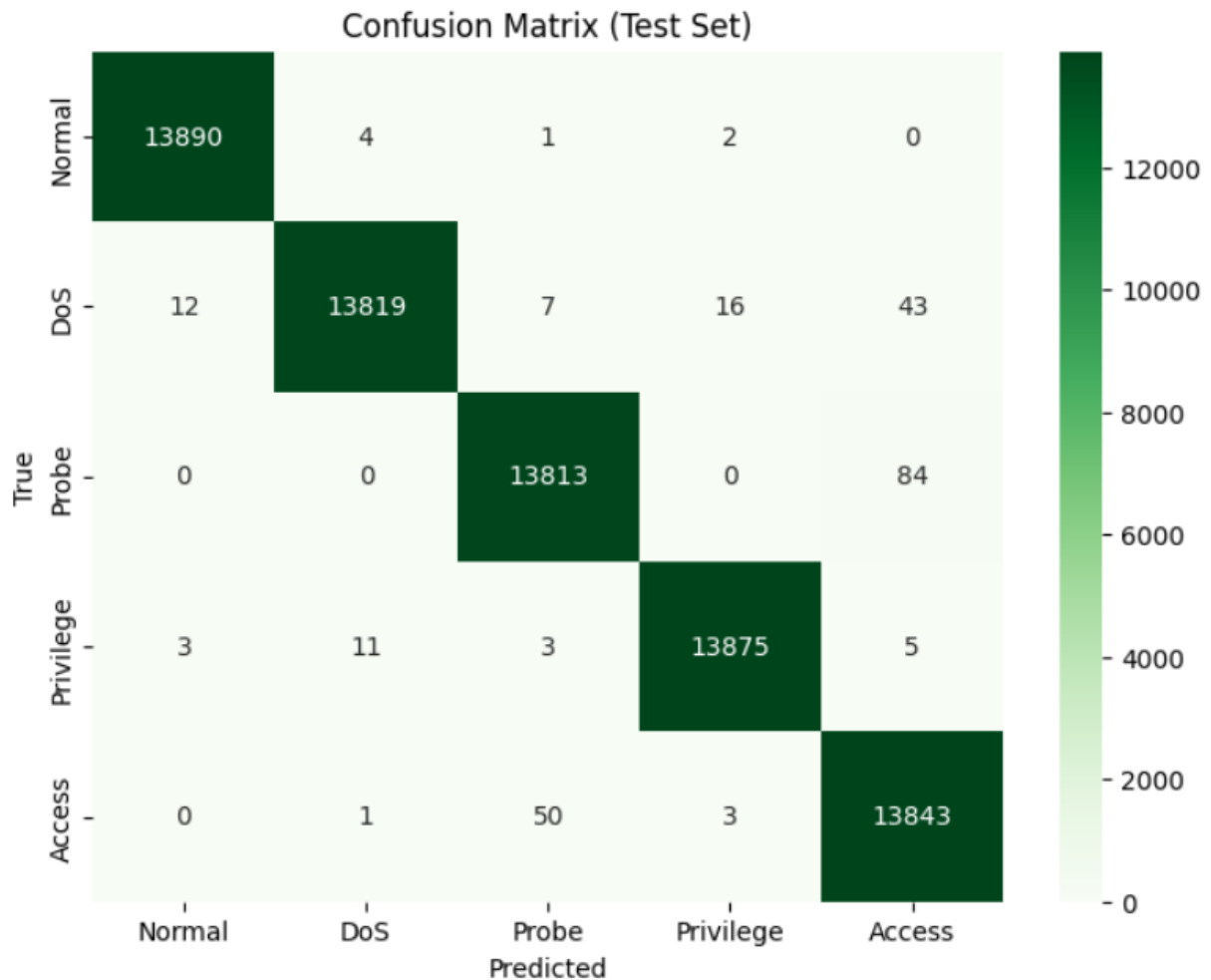




The ANN model achieved the following performance metrics:

- **Validation Loss:** 0.0147
- **Validation Accuracy:** 99.62%
- **Test Loss:** 0.0149
- **Test Accuracy:** 99.65%

The confusion matrix shows that model performs very well at predicting the classes with very little classification error, and majority of the values being concentrated on the diagonal. The visualizations demonstrate stable learning of the model on the training data and a corresponding increase in the validation accuracy.



Discussion

The high accuracy values of both validation and test data sets demonstrate the potential of an ANN-based model in classifying attacks within the NSL-KDD dataset with minimal errors. Moreover, the application of SMOTE to address the data imbalance ensured that the model didn't overfit the majority class. The confusion matrix further validates the model's ability to learn the patterns for different types of attack, and shows that misclassifications are minimal. While the model achieves a very high accuracy, it's important to remember that this is a simulation-based performance. The actual model performance on real-world data can differ, owing to the variety in real network traffic and the dynamic nature of cyber attacks.

Conclusion

This project has successfully demonstrated the application of machine learning, specifically ANNs, for network intrusion detection using the NSL-KDD dataset. Through careful data preprocessing and feature engineering, the ANN model achieved excellent accuracy in classifying network traffic into various attack categories. The visualization of the confusion matrix, and training and validation accuracy/loss curves has helped to gain further understanding of the model performance. These results further support the use of deep learning algorithms in building efficient network security solutions. Further improvements can be considered to make model even more adaptive to real-world scenarios.

Future Work:

Future work could explore:

- The impact of different network architectures or model hyperparameters on the prediction performance.
- Testing other more advanced techniques to address data imbalance.
- Implement the model on real time data feeds, and evaluate the performance.
- Explore other complex and novel attacks to see if model is able to perform on them as well.

References

1. Tavallaee, M., Bagheri, E., Lu, W., & Ghorbani, A. A. (2010). A detailed analysis of the KDD CUP 99 data set. In *2010 IEEE symposium on computational intelligence for security and defense applications* (pp. 1-6). IEEE.
2. Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: synthetic minority over-sampling technique. *Journal of artificial intelligence research*, 16, 321-357.
3. Chollet, F. (2018). Deep learning with Python. Manning Publications.

4. S. Revathi, Dr. A. Malathi, "A Detailed Analysis on NSL-KDD Dataset Using Various Machine Learning Techniques for Intrusion Detection", *International Journal of Engineering Research & Technology (IJERT)*, ISSN: 2278-0181, Vol. 2 Issue 12, December - 2013
5. Vipin Kumar, Himadri Chauhan, Dheeraj Panwar, "K-Means Clustering Approach to Analyze NSL-KDD Intrusion Detection Dataset", *International Journal of Soft Computing and Engineering (IJSCE)* ISSN: 2231-2307, Volume-3, Issue-4, September 2013
6. Santosh Kumar Sahu Sauravranjan Sarangi Sanjaya Kumar Jena, "A Detail Analysis on Intrusion Detection Datasets", *2014 IEEE International Advance Computing Conference (IACC)*